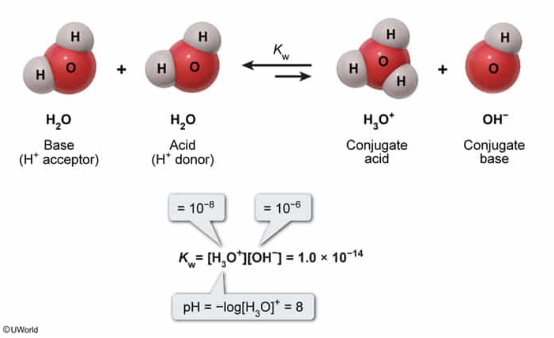
If the hydroxide concentration in common bile duct is 1×10^{-6} M, what is the pH of the bile?

- A. 6 [17%]
- B. 7 [1%]
- ✔ C. 8 [80%]
- D. 14

Correct
79% answered correctly

Explanation:



The pH of a solution relates directly to the hydrogen ion (or hydronium) concentration $[H_3O^+]$ by the relationship $pH = -\log[H_3O^+]$. If the hydroxide concentration $[OH^-]$ is known instead, it is related to the $[H_3O^+]$ by the **self-ionization constant** K_w of water as given by:

$$\mathbf{K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}\ M^2}$$

Accordingly, dividing K_w by the known $[OH^-]$ permits the calculation of $[H_3O^+]$ as follows:

$$[H_3O^+] = K_w / [OH^-] = (1.0 \times 10^{-14}\ M^2)/(1.0 \times 10^{-6}\ M)$$

$$[H_3O^+] = 1.0 \times 10^{-8}\ M$$

The pH of the solution can then be determined:

$$pH = -\log[H_3O^+] = -\log(1.0 \times 10^{-8}) = 8$$

Alternatively, the pOH can be used to find the pH, given that **$pOH = -\log[OH^-]$** and that **pH + pOH = 14** (as derived from K_w).

By this method, calculation of the pOH is given by:

$$pOH = -\log[OH^-] = -\log(1.0 \times 10^{-6}) = 6$$

and the subsequent evaluation of the pH is given by:

$$pH = (14 - pOH) = (14 - 6) = 8$$

(Choice A) A value of 6 corresponds to the pOH rather than to the pH. The calculation of $pOH = -\log[OH^-]$ whereas $pH = -\log[H_3O^+]$.

(Choice B) Based on the ionization constant K_w, the product of $[OH^-]$ and $[H^+]$ is always equal to 10^{-14}. A neutral pH of 7 requires that $[OH^-] = [H_3O^+] = 1.0 \times 10^{-7}$ M. This is not the case for the bile, which has $[OH^-] = 1.0 \times 10^{-6}$ M.

(Choice D) The value of $pK_w = 14$. This is not equivalent to the pH.

Educational objective:

The $[H_3O^+]$ and $[OH^-]$ concentrations are related by the self-ionization constant K_w of water according to the expression $K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}\ M^2$. Using a logarithmic approach, pH is related to $[OH^-]$ by the relationship $pH + pOH = 14$, where $pOH = -\log[OH^-]$.

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